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Deposition of ethyl cellulose-based drug-carrying coating by low-energy electron beam dispersion and its antifungal properties

Zhenggang Li [a], Xiaoxue Tan [a], M.A. Yarmolenko [a,b], Keneshbekova A [a], Ao Wang [a], Xin Liu [c], Xiaohong Jiang [a,*]

[a] International Chinese-Belorussian Scientific Laboratory on Vacuum-Plasma Technology, Nanjing University of Science and Technology, 200, Xiaolingwei Street, Nanjing, 210094, China
[b] Francisk Skorina Gomel State University, 104, Sovetskaya Street, Gomel, 246019, Belarus
[c] The Affiliated Hospital of Nanjing University of Chinese Medicine, Department of Orthopedics, Nanjing, 210029, China

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ABSTRACT

In this study, an EC-based dual drug-carrying composite film with ethyl cellulose (EC): silver nitrate = 1: 1 as the upper layer and EC: fluconazole (FCZ) = 1: 1 as the lower layer was prepared using the low-energy electron beam dispersion (EBD) technique. The film molecular structure, chemical composition and morphology were studied by FT-IR, XPS, and SEM methods. The results showed that the EC powder and FCZ powder used had excellent film-forming ability and the prepared films retained the majority of the molecular structure of the original powder. The films exhibited no obvious defects on the surface and their thickness reached the micrometer scale. The contact angle test confirmed good hydrophilicity of the films. With immersion in phosphate buffered saline (PBS), the porous structure formed on the surface of the films facilitated drug release. The kinetic properties of FCZ release from the composite layer into the aqueous environment were investigated, and a long-lasting release of FCZ in the composite layer for about 9 days was tailed. The performance of the films was characterized and the results showed that the EC-based dual drug-carrying composite film exhibited optimal inhibition of *Candida albicans*.

1. Introduction

Chronic wounds that are difficult to heal are at risk for fungal infections, and the moist, warm, nutrient-rich environment of the wound bed provides ideal conditions for fungal growth, making fungal infections one of the greatest health challenges. Conventional antifungal drug delivery is usually done orally, by injection, spray, or by applying ointment [1]. Although it can be therapeutically effective, it has many shortcomings, such as poor drug utilization due to systemic administration, and it may also lead to short-term overdose and long-term underdose [2,3]. Approximately 1.7 million people worldwide are reported to suffer from fungal infections, of which localized fungal infections are more common and harder to eradicate [4–6]. Despite the increasing number of fungal infections associated with biomaterials and the threat to human life, the number of studies on antifungal materials and surfaces remains limited. To overcome this deficiency, researchers have explored novel drug delivery modalities such as loading antifungal drugs into nanosponges [1,7], nanoparticles [8,9], hydrogels [10,11] and films [12,13]. Among these, films have been emphasized for their design flexibility and ability to sustain drug delivery. To the best of our knowledge, there are no relevant reports on using PVD (physical vapor deposition) methods to prepare antifungal coatings for the treatment of fungal infections. Therefore, to address these urgent needs for new therapeutic strategies to treat localized fungal infections and diseases, in this work, a novel drug sustained-release coating containing fluconazole, silver nitrate, and ethylcellulose was designed and fabricated using a low-energy electron beam dispersion (EBD) method.

Fluconazole (FCZ) is a widely used triazole antifungal agent with broad-spectrum antifungal action for the treatment of systemic or non-systemic fungal infections caused by Candida and Cryptococcus [14]. It weakens fungal cell membranes, disrupts osmotic homeostasis and inhibits fungal growth by selective inhibiting 14α-demethylase, an enzyme involved in the biosynthesis of ergosterol in fungal cells, especially against infections caused by Candida spp. and the new Cryptococcus spp [15–17].

Silver nanoparticles (AgNPs) have received much attention as a

* Corresponding author.
E-mail address: jiangxh24@njust.edu.cn (X. Jiang).

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potential antifungal complement to antibiotics, as they can bridge the gap where antifungals are usually inadequate. They were shown to have adequate antifungal properties [18,19]. In addition, the combined use of AgNPs and FCZ was found to have a synergistic effect on fungal cells [20,21], which may potentially treat local wound fungal infections.

Ethyl cellulose (EC) is a derivative of cellulose, which is produced by replacing the hydroxyl group of cellulose repeating glucose with an ethoxy group [22]. The properties of EC, such as non-toxicity [23], stability [24], biodegradability [25] and film-forming properties [26,27], make it an ideal drug carrier for sustained release applications [23]. Therefore, EC can be used to load drugs to better reduce the risk of infection during wound healing.

Low-energy electron beam dispersion (EBD) is a PVD method that performs fast deposition and easy handling, with electron beam dispersion favoring uniform high-purity and high-density films [28]. The method does not require the use of liquid media, and any material, including metals and polymerized organic compounds, can be used as a substrate. Most importantly, it does not affect the shape and strength of the substrate itself. This is because the substrate is unaffected by heat during the deposition process [29,30].

In this study, EBD technology is used to convert antifungal drugs into coatings and ethylcellulose as a carrier to design a slow-release composite film of drugs with EC and $AgNO_3$ on the top layer and EC and FCZ on the bottom layer, delivering FCZ and AgNPs for topical medication to reach sustained release of the drugs that provides a new strategy for the treatment of localized fungal infections.

2. Materials and methods

2.1. Materials

Fluconazole (FCZ, CAS 86386-73-4, 99 %) was purchased from Shanghai Yuanye Biotechnology Co., Ltd (Shanghai, China), silver nitrate ($AgNO_3$, CAS 7761-88-8, 99.8 %) was purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China), and ethyl cellulose (EC, CAS 9004-57-3) was provided by Gomel State University, Belarus. Silicon chip was purchased from Suzhou Jingsi Electronic Technology Co., Ltd (Suzhou, China). Titanium sheet was purchased from Wuxi Thousand Hammer Metal Products Co., Ltd. (Jiangsu, China). Potassium bromide (KBr) tablets were purchased from Zigong Sanchuan Industry Co., Ltd. PBS buffer was purchased from Phygene Biotechnology Co., Ltd (Fujian, China). Sabouraud Dextrose Agar (SDA) medium was purchased from Wuhu Meinuo Biological Engineering Co.

2.2. Low-energy electron beam dispersion technology

The low-energy electron beam dispersion equipment used in this work was developed independently by the Sino-Baik Vacuum Plasma Technology Science Laboratory. Before deposition, the target material is placed into the vacuum chamber, and, when the vacuum level of the equipment reaches 5×10^{-4} Pa, the electron gun is turned on, and the tungsten filament in the electron gun is heated to emit hot electrons. The hot electrons are accelerated to gain kinetic energy and bombard the target, converting the kinetic energy into heat and vaporizing to form an active gas phase [31,32]. The target was continuously irradiated until it was completely evaporated to form a film of controlled thickness on the substrate. During the deposition process, the operating current and voltage parameters fluctuated within specific range: accelerating voltage = 1.0 ± 0.5 kV, cathodic current = 0.9 ± 0.2 A. In Fig. 1 is a schematic diagram of film deposition using the EBD method. After the complete target sputtering, the power of the electron gun was switched off and, after 15 min cooling, the vacuum chamber was opened to remove the prepared samples.

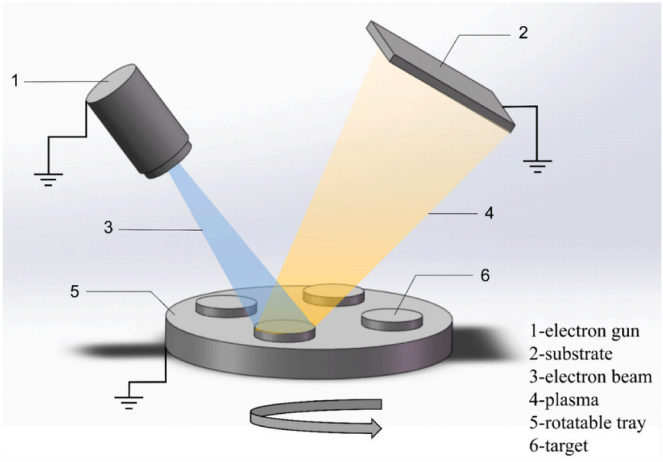


Fig. 1. Scheme of low-energy electron beam sputtering set up.

2.3. Film preparation process

2.3.1. Preparation of substrate

In this study, titanium wafers, silicon wafers and KBr salt wafers were used as substrates for various characterization methods. Before deposition, the titanium and silicon wafers were ultrasonically cleaned, first with water for 15 min, then with ethanol for 15 min, and the cycle was repeated three times before drying in an oven at 70 °C. The KBr salt flakes could be used directly.

2.3.2. Preparation of one-component films

Fluconazole powder (0.1, 0.3 and 0.6g) or $AgNO_3$ powder (0.1, 0.3 and 0.6g) was weighed with an analytical balance, put into a crucible and pressed with a crucible rod to prevent powder splattering during film deposition. Then, the crucible containing powder (powder target) was placed into a target tray in a vacuum chamber, and the EBD method was used to deposit the batch on different substrates to obtain monolayer films of fluconazole (FCZ) and $AgNO_3$ with different FCZ and $AgNO_3$ contents.

2.3.3. Preparation of EC-based composite films

EC (0.3g) + FCZ (0.3g) and EC (0.3g) + $AgNO_3$ (0.3g) were weighed according to equal weight ratios. Then, the two mixed powders were put into a crucible, ground evenly, and pressed with a crucible rod to obtain the EC + FCZ target and EC + $AgNO_3$ target. Finally, the targets were placed in a vacuum chamber. The films prepared after deposition by the EBD technique were EC + FCZ monolayer composite film and EC + $AgNO_3$ monolayer composite film, respectively.

The double-layer composite film was produced in a single process cycle without decompression of the vacuum chamber. The prepared ethyl cellulose + fluconazole powder target and ethyl cellulose + silver nitrate target were placed in the vacuum chamber simultaneously, and the ethyl cellulose + fluconazole powder target was first sputtered. After the complete sputtering of the target, the target tray was rotated to allow the electron beam to irradiate the ethylcellulose + silver nitrate target. After deposition, EC + FCZ/EC + $AgNO_3$ double-layer composite film was obtained with a bottom layer consisting of ethylcellulose and fluconazole and a top layer consisting of ethylcellulose and silver nitrate. The flowchart in Fig. 2 illustrates the preparation process of EC + FCZ/EC + $AgNO_3$ bilayer composite film.

2.4. Material characterization

2.4.1. XPS

The elemental composition of the powder and films was analyzed by X-ray photoelectron spectroscopy (Phi QUANTERA II) using Al Kα as the

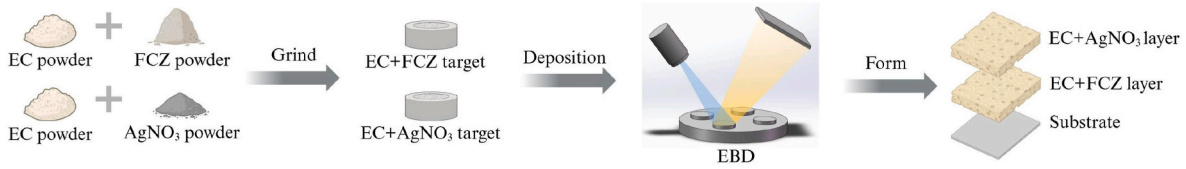


Fig. 2. The flowchart of the preparation process for EC + FCZ/EC + $AgNO_3$ bilayer composite film.

source of X-ray radiation (H = 1486.6 eV) with a spot size of 200 × 250 μm^2. The electron emission angle is 45°. Before the measurement, the sample surface was cleaned of any amorphous carbon (AdC) by argon ion etching for 1 min, with an acceleration voltage of 2 kV and an incidence angle of 45°. Analysis began once the pressure in the vacuum chamber reached 10^{-7} Pa. An integrated dual-beam charge neutralization system was used to neutralize the charge. During the analysis, the efficiency of charge neutralization was evaluated by varying the X-ray source current while keeping the C 1s peak position fixed. In view of the poor conductivity of the samples, the recommendations of the literature [33] were adopted in this paper. To more accurately calibrate the binding energy (BE) scale, we referenced literature data closely matching the characteristics of our sample and used the aliphatic C1s component for calibration, setting its binding energy at 285.0 eV [34].

2.4.2. FT-IR

The presence of functional groups and the material structure of the powder and thin film samples were analyzed using a Fourier Transform Infrared Spectrometer (FT-IR, Thermo Fisher ICOLETIS10, USA) with scanning range of 4000-400 cm^{-1} and spectral resolution of 4 cm^{-1}.

2.4.3. SEM

SEM (Hitachi Regulus 8100, Japan) is used to observe the surface morphology of the film. Since the sample is not a conductive film, the sample was sprayed with gold before testing to improve its conductivity and obtain a high-definition image. Use a gold spray instrument (Quorum Technologies Ltd) to spray gold.

2.4.4. TEM

In a transmission electron microscope (TEM), an electron beam emitted by an electron gun is converged by a condenser mirror to form a spot on the sample surface. The electron beam that passes through the sample carries structural information about its interior. TEM analysis can visualize the internal structure of a thin-film sample. In this study, the morphology and size of polymer-doped metal nanoparticles were analyzed by a JEM F-200 cold-field transmission electron microscope manufactured by Japan Electronics (JOEL), with an accelerating voltage of 200 kV.

2.4.5. Contact angle test

The contact angle is used to evaluate the wettability of a liquid on a material surface. A contact angle measuring instrument (FCA2000A2) from Shanghai AFS Precision Instruments Co. Ltd. was used to characterize the contact angle, and each experiment was repeated three times, with the measurement data expressed as mean ± standard deviation (mean ± SD).

2.5. In vitro antifungal properties

Candida albicans ATCC10231 was tested using the agar diffusion method. It was inoculated on Sabouraud Dextrose Agar (SDA) medium used for fungal cultivation. The instruments and materials used were sterilized in an autoclave. At the end of sterilization, the uninoculated liquid medium was placed on an ultra-clean bench and the fungi were inoculated into 10 ml shaker tubes using a 1 μL inoculation loop and shaken well. Pipette 100 μL of the liquid into the petri dish and spread it well using the applicator stick, then place a 6 mm diameter membrane disc (titanium base) in the petri dish to ensure that the surface of the sample is covered. It should be noted that glacial acetic acid (PH = 5.6 ± 0.2) needs to be added before placing the membrane disc to create an acidic environment. Finally, the petri dish was incubated at 37 °C in a constant temperature incubator for 24h and the diameter of the zone of inhibition was measured more than three times, with the average value recorded.

2.6. UV–Vis

Optical absorption spectra were recorded using a Thermo Fisher Evolution (USA) spectrometer. The characteristic absorption peaks of the surface plasmon resonance absorption effect of metal nanoparticles in the UV–Visible spectrum can be used to describe the properties of metal nanoparticles doped in films.

To study the release kinetics of FCZ, monolayer FCZ film and EC-based bilayer film were cut into 6 mm diameter discs and immersed in 10 ml of PBS buffer solution at 37 °C, configured with different concentrations of FCZ (10, 20, 30 and 60 μg/ml) and measured at different time points (1, 3, 5, 7, 12, 24, 48, 72, 96, 120, 168 and 216h) using UV–Vis spectroscopy at 255 nm (the maximal absorbance of FCZ). The standard curve equation y = 0.00154x-0.01296 for FCZ was plotted using absorption at 255 nm (maximum absorbance of FCZ) as measured for aliquots of 4 ml of the solution at different time points (1, 3, 5, 7, 12, 24, 48, 72, 96, 120, 168 and 216 h). Pure PBS solution was used as a reference. The concentration of FCZ in the PBS solution was calculated and the FCZ release kinetic curve was plotted by determining the maximum absorption peaks of the samples immersed in the PBS solution at different times.

2.7. Statistical analysis

All data were presented as the means ± standard deviation (SD). One-way analysis of variance (ANOVA) was used for multiple comparisons and t *t*-test to determine the difference between the two groups. Statistical analysis was performed with GraphPad Prism (Version 9.0, USA). *P < 0.05 or **P < 0.01 was considered as significant.

3. Results and discussion

3.1. Morphology and structure

3.1.1. XPS

The chemical structures of the films were analyzed using XPS. The XPS results of EC powder, EC film, and EC + $AgNO_3$ film are shown in Fig. 3. The relative intensity of the C 1s and O 1s core levels in the full spectrum of EC powder demonstrated in Fig. 3a are 69.8 % and 30.2 %, respectively. Considering the molecular structure of EC compound, the C1s band was decomposed into individual peaks, and the high-resolution energy spectrum of C 1s in Fig. 3a1 can be fitted by three peak components with binding energies 284.89, 286.35 and 287.85 eV, which are correspondent to the C-C/C-H, C-O/C-OH and O-C-O/C=O units [35,36]. In the spectrum of EC film in Fig. 3b, the relative intensities of C 1s and O 1s lines are 70.4 and 29.6 %, respectively, and the high-resolution spectrum of C 1s in EC film can be fitted by four components with the binding energies 284.81, 286.27, 287.59 and 288.93 eV, which are correspondent to the C-C/C-H, C-O/C-OH, O-C-O/C=O

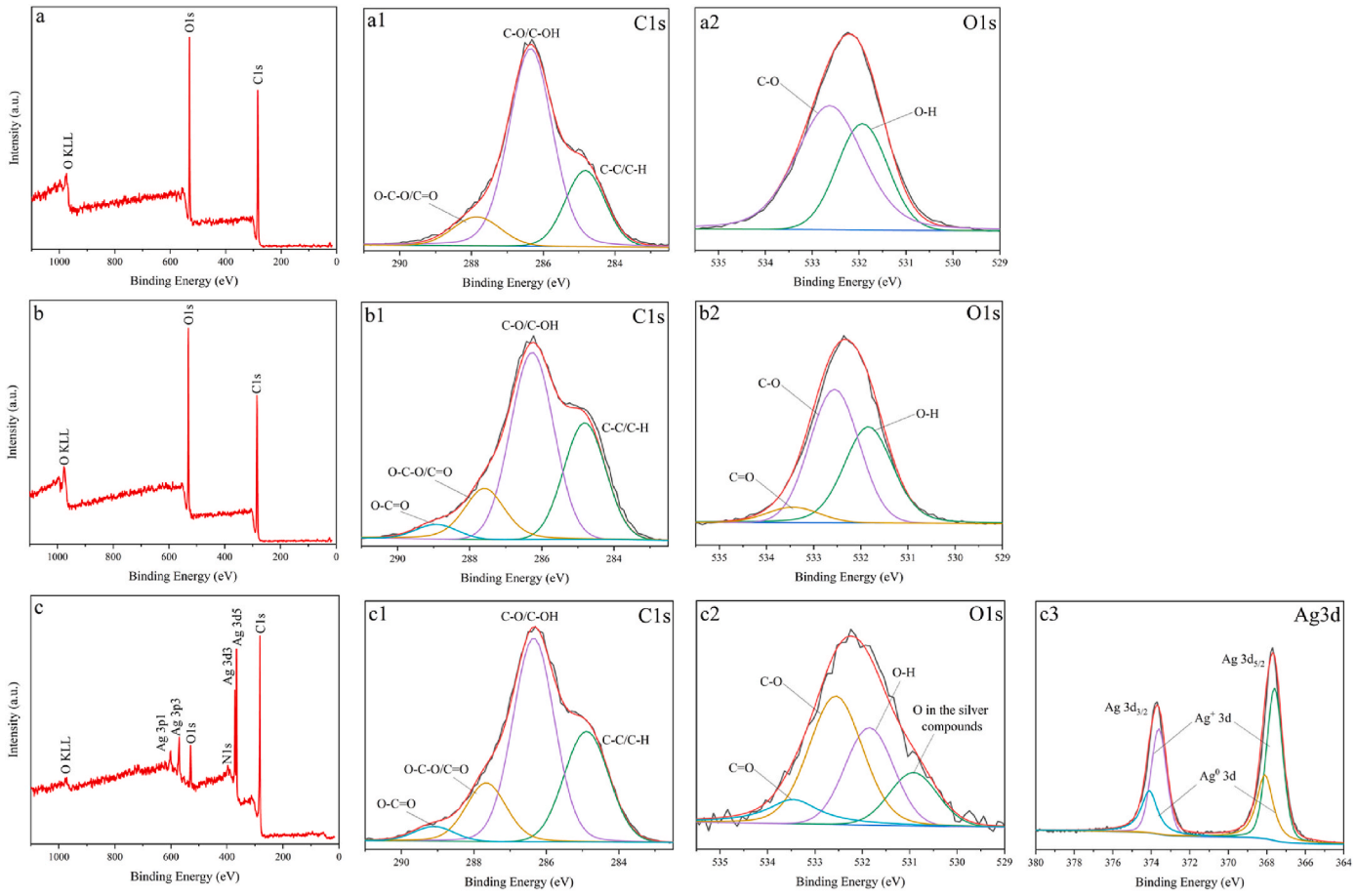


Fig. 3. XPS spectra of the films and powder: a- EC powder; b-EC film; c- EC + $AgNO_3$ film.

and O-C=O units. Compared with the spectrum of EC powder, there is an additional O-C=O at 288.93 eV, which also confirms that a small fraction of the molecular fragments were oxidized after EC film formation, as mentioned below in FTIR. In addition, as shown in Fig. 3a2, the high-resolution spectrum of O1s of EC powder can be fitted with two peaks located at 531.93 and 532.63 eV, corresponding to O-H and C-O bonds, respectively. However, the EC film (Fig. 3b2) shows a C=O bond (533.44 eV) in addition to the fitted O-H bond (531.84 eV) and C-O bond (532.56 eV), which corresponds to the high-resolution spectrum of the C1s [34]. This further proves that EC can keep most of its original structure undamaged during the low-energy electrification process and can maintain the same basic structure as the powder.

As shown in Fig. 3c, the EC + $AgNO_3$ film consists of C, O, N, and Ag elements, in which the high-resolution spectrum of C1s (Fig. 3c1), the fitted four peaks are the same as those of the single-component EC film, corresponding to 284.89, 286.35, 287.66 and 289.10 eV, respectively. As shown in Fig. 3c2, the high-resolution O1s spectrum of the EC + $AgNO_3$ film was fitted with four peaks located at 530.92, 531.85, 532.56, and 533.46 eV, corresponding to oxygen in the silver compounds within the film, as well as the O-H bond, C-O bond, and C=O bond in EC, respectively [37,38]. The XPS spectrum of the Ag 3d doublet (Fig. 3c3) has two adjacent peaks at 373.70 and 367.70 eV attributed to the Ag 3d3/2 and Ag 3d5/2 orbitals, respectively [39,40]. These two neighboring peaks can be fitted as two different sets of shoulder peaks. One set of shoulders contains two peaks at 373.61 and 367.60 eV, which correspond to the characteristic peaks of the Ag3d3/2 and Ag3d5/2 orbitals of the monovalent oxidation state Ag^+ ion, respectively. The other set of two peaks is observed at 374.10 and 368.10 eV, and the components are correspondent to the characteristic Ag 3d3/2 and Ag 3d5/2 orbitals of the zero-valent metal monomer Ag^0, respectively [41–44]. It can be seen that the XPS data results, on top of the UV–Vis spectrogram results, once again proved that $AgNO_3$ doped EC-based composite film does indeed have AgNPs, with a calculated characteristic peak area ratio of Ag^0 to Ag^+ of about 1:1.85.

In Fig. 4a is the XPS spectrum of the FCZ powder, comprising the elements C, N, F and O with relative intensities of 60.3, 23.8, 9.3 and 6.6 %, which is close to the elemental compositions of C (61.2 %), N (25 %), F (8.5 %), and O (5.3 %) of the FCZ film in Fig. 4b. The high-resolution energy spectra of C1s can be seen in Fig. 4a1 and 4b1, respectively. The components in the C1s bands are located at 285.02, 286.50, 287.40, 288.50 eV and 284.99, 286.40, 287.40, 288.49 eV respectively, corresponding to the C-C/C-H bond. C-OH bond, N-C=N bond, and F-substituted aromatic C atom on benzene ring structure in FCZ [45]. In the FCZ powder and film presented in Fig. 4a2 and 4b2, the high-resolution spectra of N1s show peaks at 397.39 and 397.60 eV. These peaks correspond to C-N, and the binding energies at 399.24 and 399.56 eV correspond to the -N = bond in the triazole ring [46–50]. Fig. 4a3 and 4b3 show the high-resolution spectra of O1s for FCZ powder and FCZ film. Both of them can be fitted with O-H and C-O bonds located at the positions of 531.61、532.47 (FCZ powder) and 531.56、532.46 eV (FCZ film), respectively [37]. In addition, in Fig. 4a4 and 4b4 are high-resolution spectra of F1s at 687.52 and 687.49 eV, respectively, corresponding to the C-F bond on the six-membered ring in FCZ [48,51]. There is no significant difference in the high-resolution spectra of C1s, N1s, and F1s of FCZ powder and FCZ film, which further proves that FCZ can maintain its original structure without being destroyed during low-energy electron beam sputtering and can keep the same basic structure as that of powder.

In Fig. 4c is the XPS spectrum of a monolayer composite film deposited on a Ti substrate by mixing FCZ powder and EC powder in

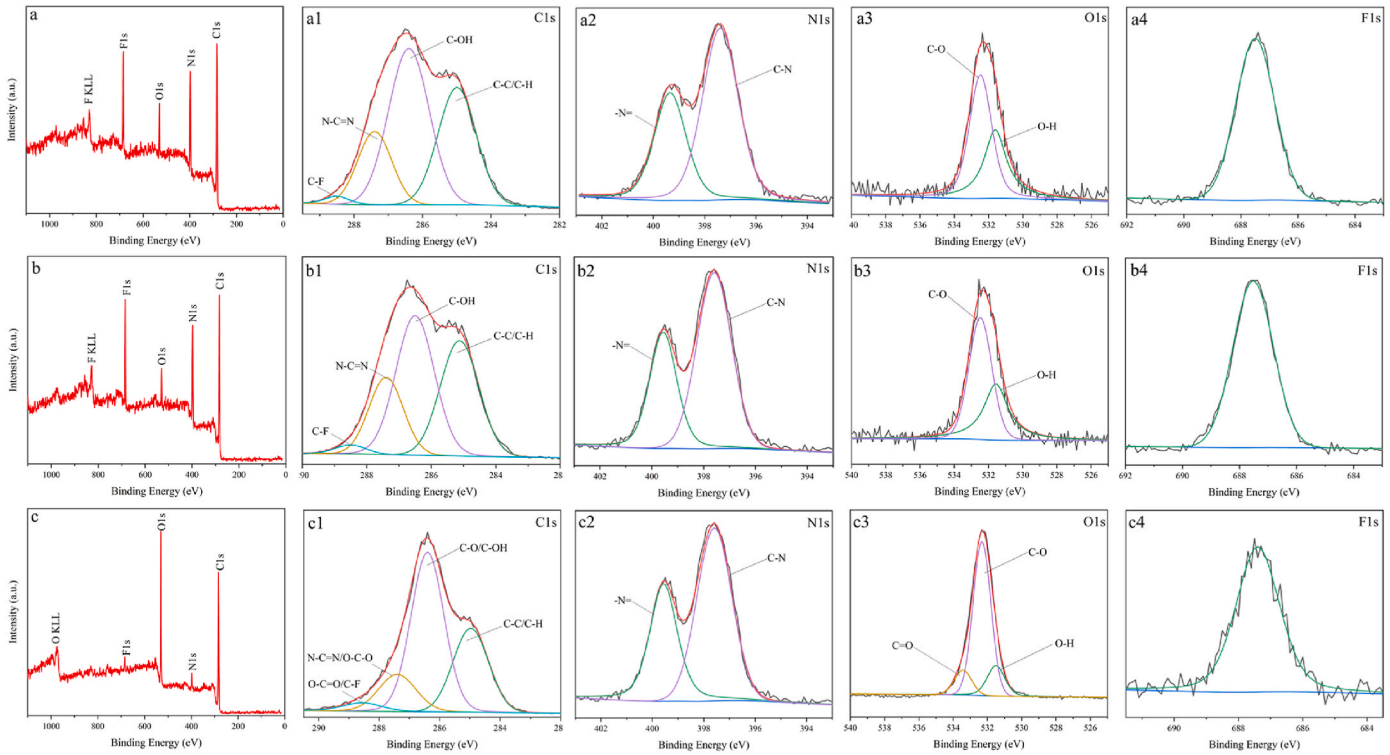


Fig. 4. XPS spectra of the films and powder: a-FCZ powder; b-FCZ film; c-EC + FCZ film.

equal mass proportions with the main components of C (67.3 %), N (4.5 %), F (2.0 %) and O (26.1 %). The increase in the contents of the C and O elements in comparison with the one-component film and powder is attributed to the predominance of C and O in cellulose. The peaks of C1s (Fig. 4c1) are located at 284.97, 286.40, 287.40, and 288.56 eV, representing the C-C/C-H bond, C-O/C-OH bond, O-C-O bond, or carbon in the triazole ring [52,53], and the O-C=O/C-F bond [36,37,45], respectively. N 1s and F1s peaked at 397.59, 399.56, and 687.40 eV, respectively (Fig. 4c2, c4), with no significant difference compared to the single-component film. The high-resolution spectrum of O1s in Fig. 4c3 can be fitted to three peaks located at positions 531.50, 532.32, and 533.43 eV, which correspond to the O-H bond, C-O bond in EC or FCZ, and C=O bond in EC, respectively [37]. These results provide evidence that FCZ was successfully mixed with EC to form film and there was no interaction between the two.

Then, the EC + FCZ/EC + EC + $AgNO_3$ film was etched (12 times every 5 min) using in situ argon ion etching to study the distribution of FCZ within the film, and the related XPS spectra are shown in Fig. 5. Since F is the characteristic element of FCZ, the distribution of FCZ molecules in the film can be estimated by the variation of the elemental content of F with the depth of the film. F1s have only one peak at 687.40 eV. In the F1s spectra, it can be seen that the F1s peak height increases as the etching depth increases. It can be inferred that FCZ diffuses unevenly from the underlying EC + FCZ film into the upper EC + $AgNO_3$ film. The closer to the bottom film, the higher the content of FCZ molecules. The thickness of EC + $AgNO_3$ film can be roughly calculated, ranging from 1 μm to 1.2 μm, which explains that no obvious interface can be observed in the SEM images.

3.1.2. FT-IR

To analyze whether the structure of the films prepared by the EBD technique has changed, the prepared powder and films were characterized in the FTIR range. As shown in Fig. 6a and b, the -CH_3 stretching vibration located at 2869-2975 cm^{-1} can be found in the FTIR spectra of both EC powder and EC film. In addition, the characteristic peak of EC powder at 1054 cm^{-1} corresponds to C-O-C stretching in cyclic ether,

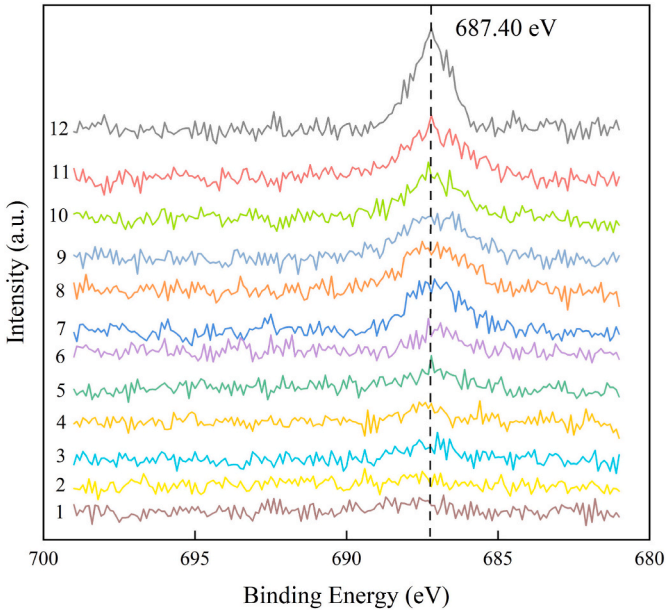


Fig. 5. XPS spectra of F1s range in the EC + FCZ/EC + $AgNO_3$ film as a function of etching time (etch for 5 min per curve).

which corresponds to 1055 cm^{-1} in the EC film [54]. Besides these bands, the low intensity bands observed at ~3500 cm^{-1} could be attributed to the presence of OH units [55,56]. Compared to EC powder, the film shows a stretching vibration of C=O at 1728 cm^{-1}, which may be caused by a large number of molecular, ionic, and plasma fragments produced when the EC powder target is bombarded by the electron beam with chemical bond breaking and reorganization, which results in a lowering of the vacuum and oxidation of some of the fragments.

The infrared spectra of FCZ powder and FCZ film are shown in Fig. 6c

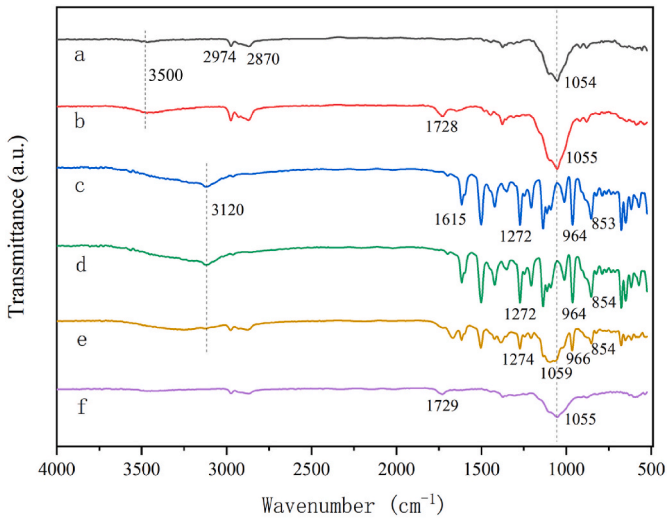


Fig. 6. FTIR spectra of the as-deposited films and original powder: a-EC powder; b-EC film; c-FCZ powder; d-FCZ film; e-EC + FCZ film; f-EC + $AgNO_3$ film.

and d. The sharp band located near 3120 cm^{-1} is caused by the aromatic C-H stretching vibration in FCZ. Also, the sharp peak near 1615 cm^{-1} in FCZ powder and FCZ film is related to the stretching of C=C and C-N bonds on the triazole ring in FCZ [57]. The absorption bands near 1272 cm^{-1} and at 964 cm^{-1} and 853 cm^{-1} correspond to those near 1272 cm^{-1} and at 964 cm^{-1} and 854 cm^{-1} of the FCZ film, which are related to the C-F stretching of FCZ and the C-H stretching of triazole ring, respectively. This indicates that FCZ retains most of its characteristic functional groups after film formation, verifying its film formation possibility.

In Fig. 6e is the infrared spectrum of EC + FCZ composite film with the C-O-C stretching vibration located at 1059 cm^{-1}, where the characteristic peaks of EC and FCZ are partially integrated due to the relative similarity and mobility of the characteristic peaks. The C-H stretching belonging to the aromatic group in FCZ and the stretching of C=C and C-N bonds on the triazole ring are also observed in the spectrum of composite film. Additionally, peaks at 1274 cm^{-1}, 964 cm^{-1} and 853 cm^{-1} are C-F stretching of FCZ with C-H stretching of the triazole ring. The relative intensities of most of the characteristic peaks were reduced compared to the single component EC and FCZ. This may be due to the formation of intermolecular hydrogen bonds between the different elements. Hydrogen bonds formed between two different macromolecules are generally stronger than those formed between molecules of the same polymer [4,58,59]. The above analyses and comparisons show that the characteristic functional groups of both EC and FCZ are present in the infrared spectrum of the composite film, which is further evidence that both were successfully evaporated and most of the functional groups of the powder are retained after film formation.

Finally, Fig. 6f shows the $AgNO_3$-doped EC film, which contains all the characteristic peaks in the FTIR spectrum of the single-component EC film, indicating that the addition of $AgNO_3$ does not change the structure of EC.

3.1.3. SEM

In Fig. 7 is the surface morphology of the dual drug-loaded film (EC + FCZ/EC + $AgNO_3$) deposited on Ti substrate and its immersion in PBS for 3 and 9 days. As seen in Fig. 7a and a1, the film surface contains some rounded spherical hills. This phenomenon is due to the tendency of the polymer film to condense into droplets during high-temperature evaporation due to its low melting point [60]. As the film was immersed in PBS, the film surface displayed a pore-like structure resulting from the slow dissolution of EC, as seen in Fig. 7b and b1, and, with the dissolution of EC, FCZs encapsulated by EC with AgNPs were released. The surface morphology of the film after 9 days of release is presented in Fig. 7c and 7c1. It can be seen that, with the film release, only some scattered fragments are distributed on the surface. It can also be seen by the EDX scanning spectrum of the bilayer film after 9 days of release (Fig. 8) that the mass fraction of Ti element on the residual film surface is 77.2 %. The Ti substrate is mostly exposed on the surface. Furthermore, Ag and F elements, characteristic of FCZ, were not detected. Only a few undegraded C elements (11.6 wt%) and O elements (11.2 wt%) remain on the surface. These indicated that the AgNPs and FCZ in the film were mostly released in about 9 days, in which the release cycle of FCZ can also be verified in the drug release kinetics study.

The monolayer EC + FCZ film and the EC + FCZ/EC + $AgNO_3$ bilayer film are shown in Fig. 9, with thicknesses of 1.41 μm and 2.47 μm,

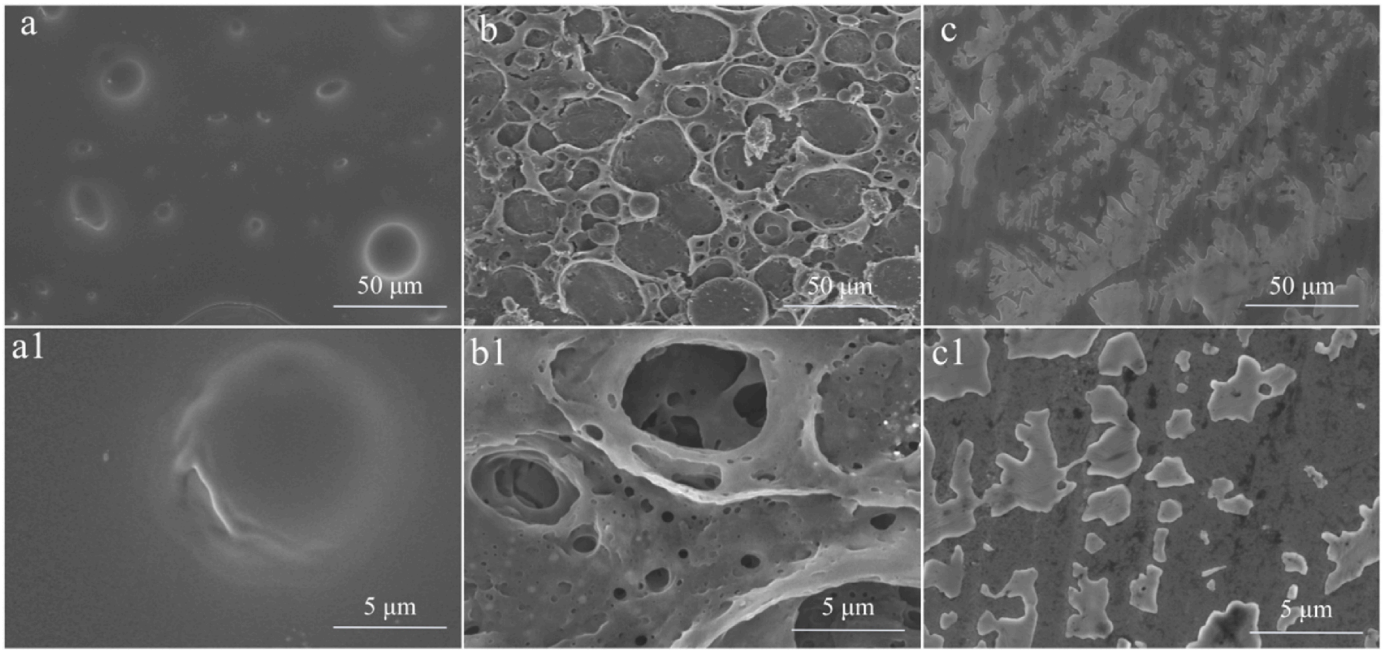


Fig. 7. SEM surface morphology of the EC + FCZ/EC + $AgNO_3$ film with different times of immersion in buffer solution: a1-0d; b, b1-3d; c, c1-9d.

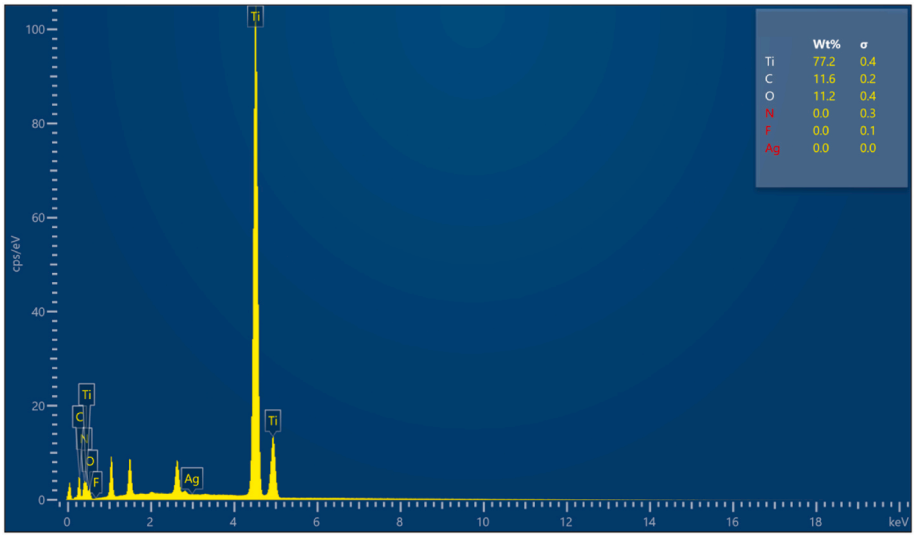


Fig. 8. EDX spectrum of the EC + FCZ/EC + $AgNO_3$ film after 9d immersion in PBS.

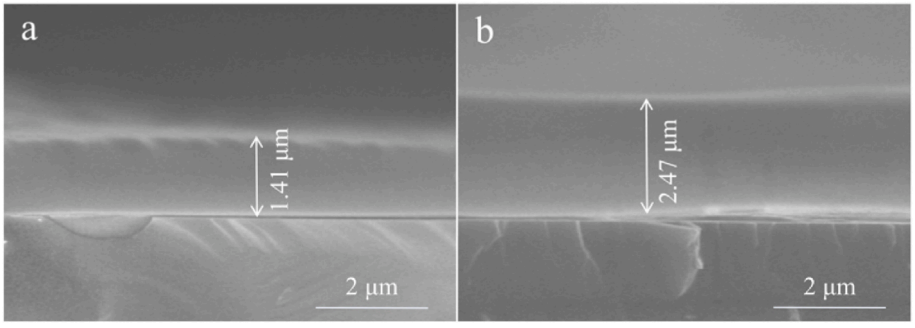


Fig. 9. Cross-sectional SEM images of the as-deposited films: a-EC + FCZ film; b-EC + FCZ/EC + $AgNO_3$ film.

respectively. For the bilayer film, the interface is blurred and there is no obvious delamination.

3.1.4. TEM

The EC + $AgNO_3$ monolayer composite film deposited directly on the substrate of the KBr salt flake was placed in distilled water. The film was detached by dissolving of the KBr salt flake, and then the detached film was dissolved in distilled water and put into an ultrasonic cleaner to achieve a uniform distribution. Subsequently, 10 μL of the solution was taken with a pipette and dripped onto the Cu mesh, air-dried at room temperature. The final sample obtained was used for the TEM test.

The TEM images of EC + $AgNO_3$ monolayer composite film obtained using the above method are shown in Fig. 10. Dark-colored spherical nanoparticles AgNPs with a particle size in the range of 30–60 nm can be seen in Fig. 10a. The lattice planes in AgNPs are well organized, as seen in Fig. 10b, with the observed lattice plane spacing of 0.235 nm, which matches well with the value of Ag(111) crystal plane of metal in the face-centered cubic crystal system (d = 0.23592 nm), verifying the presence

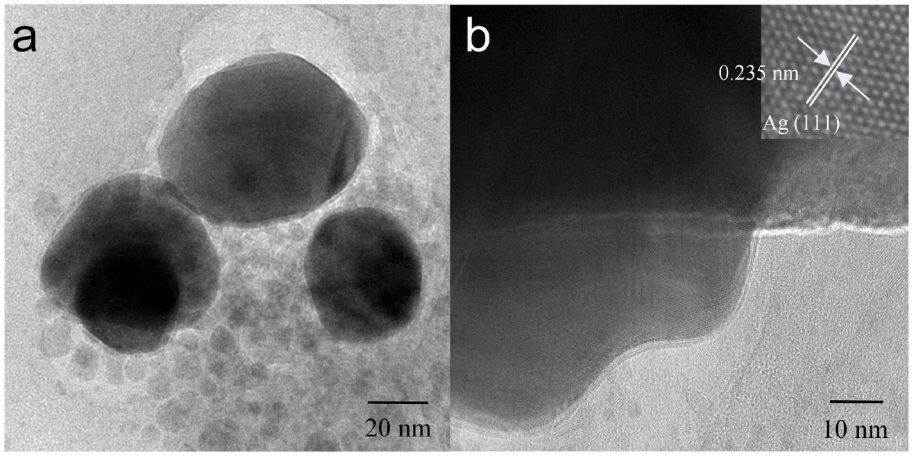


Fig. 10. TEM images of AgNPs in the treated EC + $AgNO_3$ film.

of AgNPs in the EC + $AgNO_3$ monolayer composite film.

As shown in Fig. 11 and Table 1, the elemental content of the samples was quantitatively analyzed by TEM-EDX. The EDX pattern shows the presence of C, O, K, Br, and Ag in the treated EC + $AgNO_3$ film. The content of C was 53.58 %. In addition to the composition of elemental C in EC, the material of the receiving window of the EDX detector led to the appearance of high C content. The other detected elements include K and Br, which are components of the KBr salt flake used as a substrate. Finally, the appearance of AgNPs can be observed in the EDS scans of the samples, further verifying the presence of AgNPs in the EC + $AgNO_3$ film prepared by the EBD method.

3.1.5. Contact angle

Hydrophilic surfaces are crucial for wound dressings and tissue engineering [61]. In Fig. 12 are the contact angles of different samples deposited on titanium sheets after dropwise addition for 3s. The contact angle of the pure titanium sheet (Fig. 12a), used as a control, was about 70.46°. When EC powder (Fig. 12b) converts to EC film (Fig. 12c), the water contact angle changes from 68.02° to 63.33°, indicating a hydrophilic surface in both cases. The contact angles for the monolayer EC + FCZ film (Fig. 12d) and EC + $AgNO_3$ film (Fig. 12e) were about 52.86° and 55.85°, respectively. when these two films were composited into a dual drug-carrying composite film, the aqueous contact angle of the film (Fig. 12f) was about 49.24°, which is not much different from that of the monolayer film, and both of them showed hydrophilic properties.

From the above data, it can be concluded that the EC-based films prepared by EBD technology have good hydrophilicity, which makes them more biocompatible in wound dressings and skin tissue engineering, which is more conducive to wound healing.

3.2. Study of antimicrobial properties of films

The inhibitory effect of membrane discs on *Candida albicans* ATCC10231 was studied using the disc diffusion method. *Candida albicans* is a representative fungus commonly found on skin surfaces and mucous membranes, is the most common cause of invasive fungal infections, with a mortality rate of up to 40 % [62,63], and is a good example of a pathogen to study in acquired infections.

Table 1
Relative content of elements in the sample (Wt %).

Sample	Wt (%)				
	C	O	K	Br	Ag
EC + $AgNO_3$ film after treatment	53.98	10.80	8.08	11.45	15.69

Firstly, we screened FCZ and $AgNO_3$ powder with different target material quantities for deposition to obtain monolayers of FCZ and $AgNO_3$ films, which were sensitive to the concentration of Candida albicans, as shown in Table 2 and Fig. 13a. The zone of inhibition increased from 22.67 ± 0.58 mm to 30.17 ± 0.29 mm by increasing the original target amount from 0.1 g to 0.3 g. However, by further increasing the target amount to 0.6 g, the zone of inhibition was approximately constant at about 28.23 ± 0.25 mm. This indicates that the FCZ-coated film with the original target amount of 0.3 g is the best choice for the antifungal activity of *Candida albicans*. For $AgNO_3$ coating, we used 0.1 g (12.80 ± 0.17 mm), 0.3 g (14.97 ± 0.07 mm), and 0.6 g (14.53 ± 0.31 mm) target powder for deposition, in which the antimicrobial effect of the coating with a target mass of 0.3 g against *Candida albicans* was close to that of the coating with a target mass of 0.6 g. To save the raw materials, next, we will use the coating with a target mass of 0.3 g. To save raw materials, we will use 0.3 g of FCZ powder and mixture of 0.3 g of $AgNO_3$ powder with EC powder to deposit the composite coating.

The antifungal effect against *Candida albicans* after 24 h of incubation with the addition of EC is shown in Table 2 and Fig. 13b. Pure EC discs were used as a control group. The results showed that the inhibition zones of the single-layer EC + FCZ film and single-layer EC + $AgNO_3$ film reached 28.2 ± 0.26 mm and 13.1 ± 0.26 mm, respectively. The zone of inhibition of the dual-loaded film (EC + FCZ/EC + $AgNO_3$) reached 35.07 ± 0.31 mm, which was significantly higher than that of the two single-layer EC-based films. This suggests that the simultaneous release of FCZ and AgNPs in the film provided a synergistic antifungal

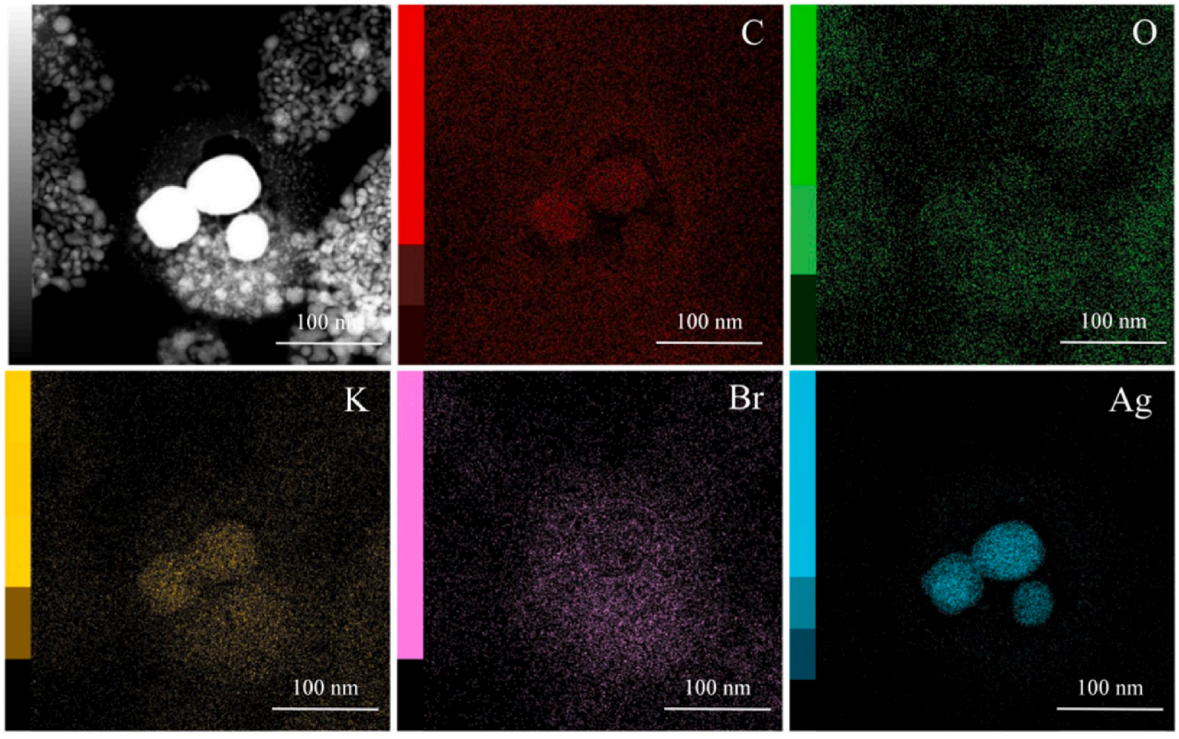


Fig. 11. TEM-EDX plots of AgNPs in the treated EC + $AgNO_3$ film.

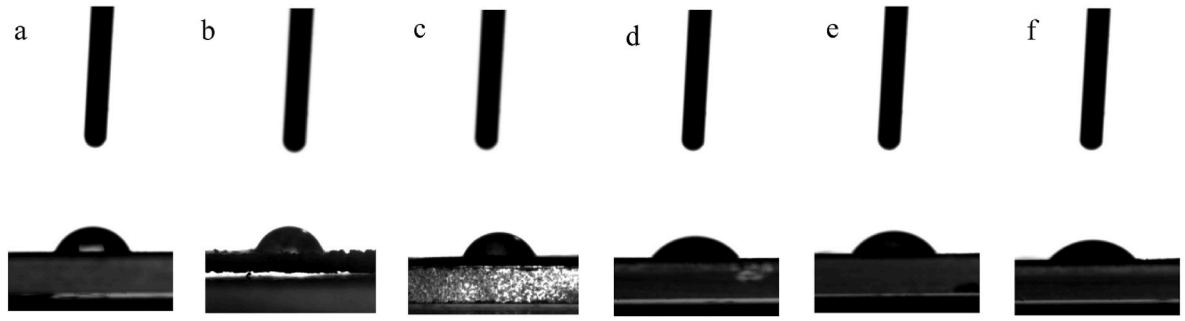


Fig. 12. Contact angles of films: a-Ti substrate, b-EC powder, c-EC film, d-EC + FCZ film, e-EC + $AgNO_3$ film, f-EC + FCZ/EC + $AgNO_3$ film.

Table 2
Inhibition zone of *Candida albicans* by FCZ film and $AgNO_3$ film prepared from different masses of target powder and EC + FCZ, EC + $AgNO_3$, EC + FCZ/EC + $AgNO_3$, FCZ + $AgNO_3$ film, blank Ti substrate (mm).

Types of films		*C.albicans 10231*	
		Inhibition zone	Standard deviation
FCZ	0.1g	22.67	0.58
	0.3g	30.17	0.29
	0.9g	28.23	0.25
$AgNO_3$	0.1g	12.80	0.17
	0.3g	14.97	0.07
	0.9g	14.53	0.31
EC + FCZ		28.20	0.26
EC + $AgNO_3$		13.10	0.26
EC + FCZ/EC + $AgNO_3$		35.07	0.31
FCZ + $AgNO_3$		41.30	0.14
Ti substrate		0.00	0.00

effect. Additionally, the antifungal property of the film prepared by mixing 0.3g FCZ and 0.3g $AgNO_3$ was tested. The diameter of the inhibition zone was about 41.30 mm, which was significantly higher than that of the EC + FCZ/EC + $AgNO_3$ film, indicating that the EC had a certain delaying effect on the release of FCZ and AgNPs.

3.3. UV–Vis

3.3.1. Characterisation of AgNPs

The UV–Vis spectra of $AgNO_3$ film samples and EC + $AgNO_3$ composite film samples are shown in Fig. 14. It can be seen that the UV–Vis spectra of both pure $AgNO_3$ film and EC + $AgNO_3$ composite film show an absorption peak around 440 nm, which is the surface plasmon resonance absorption peak of AgNPs [64–66]. This indicates that AgNPs were present in both the $AgNO_3$ film samples and EC + $AgNO_3$ composite film samples. The results also show that AgNPs with narrower particle sizes were successfully doped into the EC polymer matrix film during the deposition of EC + $AgNO_3$ composite film using the EBD method, as verified by the TEM results above.

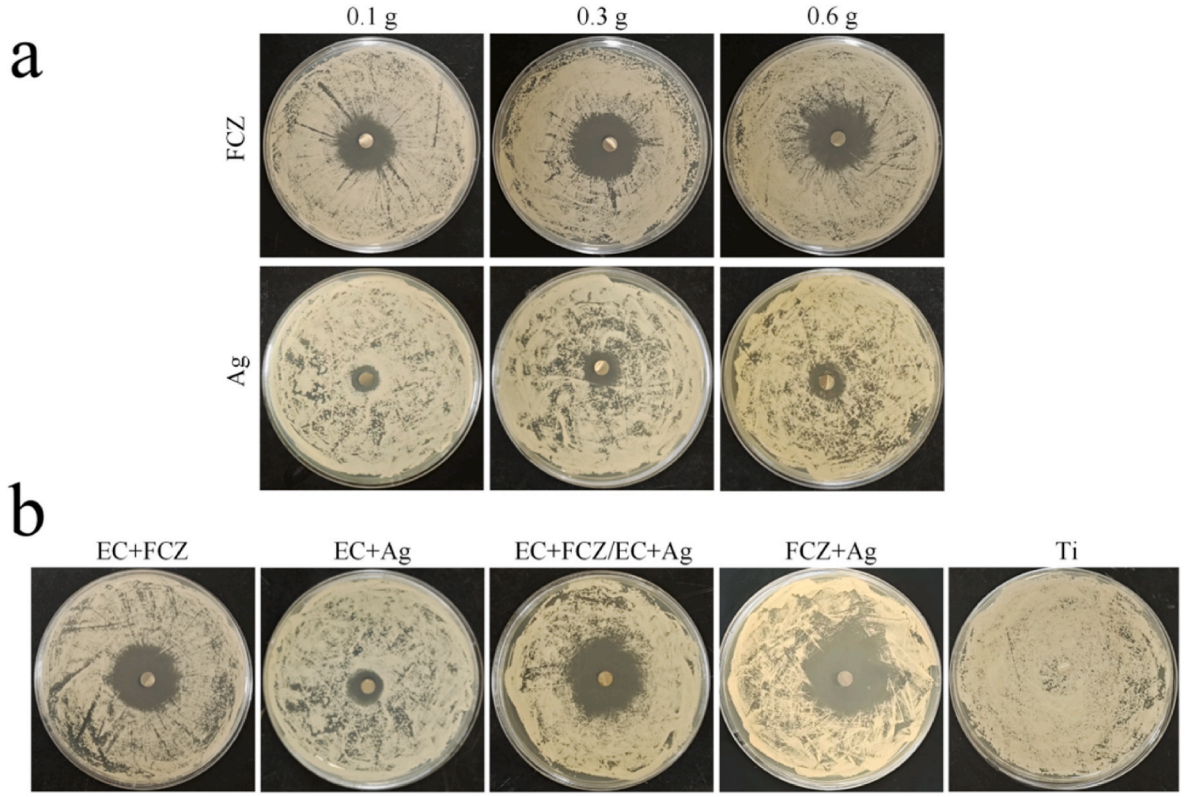


Fig. 13. a-Activity against *Candida albicans* of films prepared with 0.1 g, 0.3 g, 0.9 g FCZ target, and $AgNO_3$ target. b-Activity of EC + FCZ, EC + $AgNO_3$, EC + FCZ/EC + $AgNO_3$, FCZ + $AgNO_3$ and blank Ti substrate against *Candida albicans*.

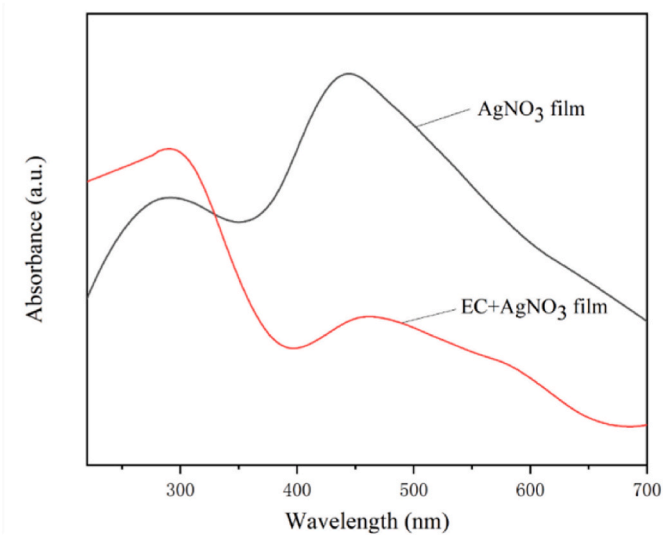


Fig. 14. UV–Vis spectra of $AgNO_3$ film and EC + $AgNO_3$ two-component film.

3.3.2. Drug release

Samples of monolayer FCZ film and EC + FCZ/EC + $AgNO_3$ dual drug-carrying composite film with a diameter of 6 mm were immersed in 10 ml of PBS and kept in a thermostatic water bath at 37 °C. The concentration-absorbance standard curve for fluconazole is shown in Fig. 15a. The absorbance of the solution at 255 nm (Fig. 15b) was measured by an UV–Vis spectrophotometer to obtain the release kinetics of FCZ from FCZ film and EC + FCZ/EC + $AgNO_3$ film. From the release curve of FCZ monolayer film (Fig. 15c), it can be seen that the FCZ concentration stabilized after about 96 h. From the release curve of the bilayer film (Fig. 15d), it can be noted that rapid release occurred within the first 36 h, after which the EC in the composite layer gradually dissolved into pores, facilitating further FCZ release. The release rate then slowed, leveling off around 216 h. Recalling Nader, J.'s previous study on polymer-loaded fluconazole release kinetics, FCZ that lasted at least 48 h was released into the surrounding medium [12]. In contrast, this study extended the release cycle of FCZ even further while avoiding the liquid phase method. The results showed that even with the direct interaction with PBS, the release period of the membrane could reach about 216 h, which greatly delayed the release of FCZ in the membrane compared with the FCZ monolayer membrane, thus ensuring its long-term therapeutic efficacy.

4. Conclusion

In summary, EC successfully carried AgNPs and undamaged FCZ into the composite film by the EBD technique, which can be regarded as highly dispersed mechanical mixtures of the initial compounds. XPS and FT-IR analyses confirmed that the EC film and FCZ film as well as the EC-based composite film prepared using the EBD technique showed no significant chemical changes compared to the original powder. The XPS etching results showed that the FCZ was distributed throughout the EC matrix by diffusion or dispersion, and the vertically distributed content was gradually increased from the surface towards the interfacial layer. The films exhibited good hydrophilicity, and their surface is continuous without obvious defects. The thicknesses of the monolayer and bilayer films were 1.41 and 2.47 μm, respectively, and there was no obvious interface between the EC + $AgNO_3$ and the EC + FCZ layers. The dual

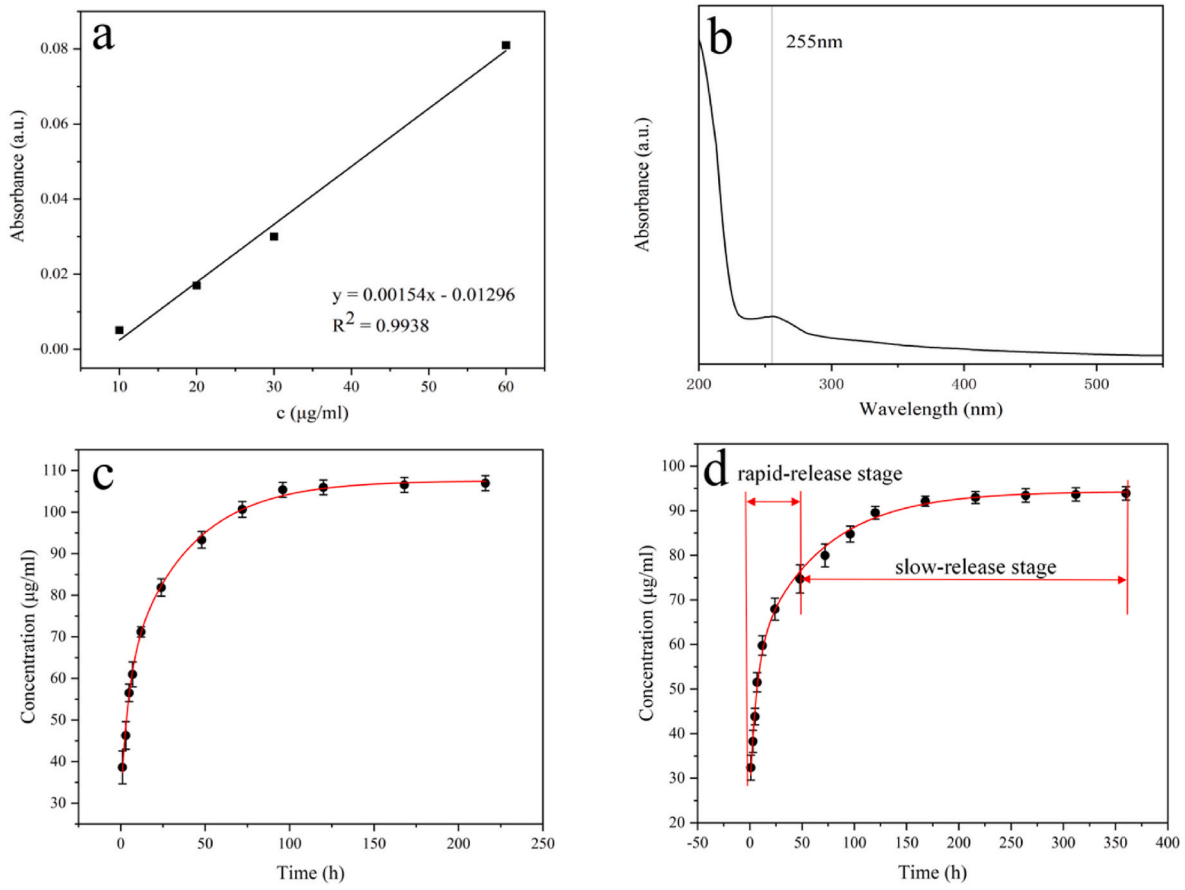


Fig. 15. a-The concentration-absorbance standard curve for fluconazole, b-UV absorption spectrum of fluconazole, c-Dynamic release curve of FCZ film in PBS, d-Dynamic release curve of EC + FCZ/EC + $AgNO_3$ film in PBS.

drug-carrying composite film released the drug through the pore structure formed by the degradation of EC, and the release of FCZ can be divided into a rapid-release phase (0–36 h) and a slow-release phase (36 h-9 d), which greatly delayed the drug release, as compared with the single-layer FCZ film. The films were tested for their antifungal properties, and the results showed that the films are effective against *Candida albicans*, The combined antifungal capacity of the bilayer film is more significant.

CRediT authorship contribution statement

Zhenggang Li: Writing – original draft, Data curation. **Xiaoxue Tan:** Data curation. **M.A. Yarmolenko:** Data curation. **Keneshbekova A:** Data curation. **Ao Wang:** Data curation. **Xin Liu:** Writing – review & editing. **Xiaohong Jiang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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